



## Why data pipelines on cloud are cheaper than on-prem solution? Grow **Data** Skills

Data pipelines on the cloud are often cheaper than on-premises solutions due to several reasons:

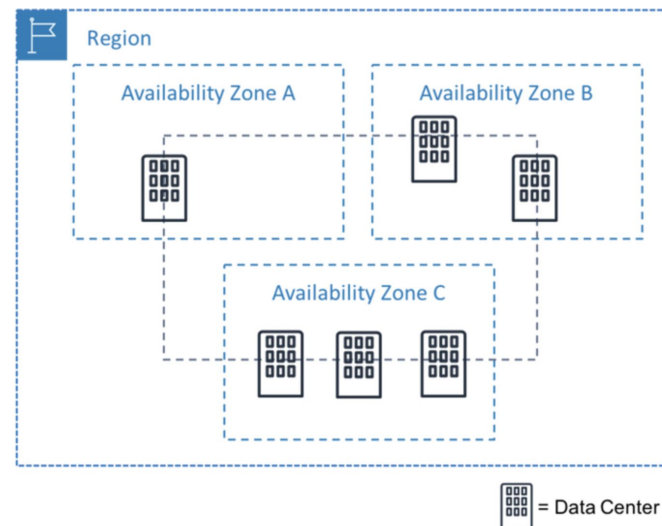
- **Infrastructure Costs:** Cloud platforms eliminate the need to invest in physical hardware, reducing both upfront capital expenses and ongoing maintenance costs.
- **Scalability:** Cloud services allow for on-demand scalability. Instead of over-provisioning resources, businesses can scale up or down based on actual usage, optimizing costs.
- **Operational Efficiency:** Cloud providers handle much of the routine maintenance, patching, and updates, reducing the manpower required for these tasks on-premises.
- **Resource Utilization:** Cloud platforms typically offer better resource utilization and multi-tenancy benefits, allowing users to only pay for the resources they consume.
- **Reduced Overhead:** On-premises solutions often require dedicated IT teams for setup, maintenance, and troubleshooting. Cloud solutions can reduce or eliminate these overhead costs.
- **Flexibility:** The cloud offers various pricing models, such as pay-as-you-go, reserved instances, or spot instances, enabling cost optimizations based on workload patterns.
- **Innovation Pace:** Cloud providers continuously introduce new features, tools, and integrations, often at no additional cost. This can lead to more efficient and cost-effective pipeline designs over time.
- **Disaster Recovery & Redundancy:** Implementing redundancy and disaster recovery on-premises can be expensive. Cloud providers often offer these features at a fraction of the cost.

## Common Services by AWS - Azure - GCP

| <b>CLOUD COMPARISON</b><br><b>AZURE VS. AWS VS. GOOGLE COMPUTE</b> |                                                                                                                |                                                                                                                                |                                                                                                                                  |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                                                                    |                              |                                             |                                               |
| <b>Available Regions</b>                                           | Azure Regions                                                                                                  | AWS Regions and Zones                                                                                                          | Google Compute Regions & Zones                                                                                                   |
| <b>Compute Services</b>                                            |  Virtual Machines             |  Elastic Compute Cloud (EC2)                |  Compute Engine                               |
| <b>App Hosting</b>                                                 |  Azure Cloud Services         |  Amazon Elastic Beanstalk                   |  Google App Engine                            |
| <b>Serverless Computing</b>                                        |  Azure Functions              |  AWS Lambda                                 |  Google Cloud Functions                       |
| <b>Container Support</b>                                           |  Azure Container Service      |  EC2 Container Service                      |  Container Engine                             |
| <b>Scaling Options</b>                                             |  Azure Autoscale              |  Auto Scaling                               |  Autoscaler                                   |
| <b>Object Storage</b>                                              |  Azure Blob Storage           |  Amazon Simple Storage (S3)                 |  Cloud Storage                                |
| <b>Block Storage</b>                                               |  Azure Managed Storage        |  Amazon Elastic Block Storage               |  Persistent Disk                              |
| <b>Content Delivery Network (CDN)</b>                              |  Azure CDN                    |  Amazon CloudFront                          |  Cloud CDN                                    |
| <b>SQL Database Options</b>                                        |  Azure SQL Database           |  Amazon RDS                                 |  Cloud SQL                                    |
| <b>NoSQL Database Options</b>                                      |  Azure DocumentDB             |  AWS DynamoDB                               |  Cloud Datastore                              |
| <b>Virtual Network</b>                                             |  Azure Virtual Network        |  Amazon VPC                                 |  Cloud Virtual Network                        |
| <b>Private Connectivity</b>                                        |  Azure Express Route          |  AWS Direct Connect                         |  Cloud Interconnect                           |
| <b>DNS Services</b>                                                |  Azure Traffic Manager        |  Amazon Route 53                            |  Cloud DNS                                    |
| <b>Log Monitoring</b>                                              |  Azure Operational Insights |  Amazon CloudTrail                        |  Cloud Logging                              |
| <b>Performance Monitoring</b>                                      |  Azure Application Insights |  Amazon CloudWatch                        |  Stackdriver Monitoring                     |
| <b>Administration and Security</b>                                 |  Azure Active Directory     |  AWS Identity and Access Management (IAM) |  Cloud Identity and Access Management (IAM) |
| <b>Compliance</b>                                                  |  Azure Trust Center         |  AWS CloudHSM                             |  Google Cloud Platform Security             |
| <b>Analytics</b>                                                   |  Azure Stream Analytics     |  Amazon Kinesis                           |  Cloud Dataflow                             |
| <b>Automation</b>                                                  |  Azure Automation           |  AWS Opsworks                             |  Compute Engine Management                  |
| <b>Management Services &amp; Options</b>                           |  Azure Resource Manager     |  Amazon CloudFormation                    |  Cloud Deployment Manager                   |
| <b>Notifications</b>                                               |  Azure Notification Hub     |  Amazon Simple Notification Service (SNS) | None                                                                                                                             |
| <b>Load Balancing</b>                                              |  Load Balancing for Azure   |  Elastic Load Balancing                   |  Cloud Load Balancing                       |

## AWS Region, Availability Zone and Data Centers

- **Region:** An AWS Region is a geographic area that contains two or more Availability Zones. AWS Regions are completely independent of one another and are designed to be completely isolated from each other, to ensure the highest levels of data privacy and resilience. A region represents a large area, such as the US West (Oregon) or Europe (Ireland).
- **Availability Zone:** An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. Each AZ is designed to be isolated from failures in other AZs and to provide inexpensive, low-latency network connectivity to other AZs in the same region.
- **Data Center:** A physical facility that houses the servers and networking equipment necessary to provide the various AWS services. Multiple data centers make up an Availability Zone, and multiple Availability Zones make up a Region.



## AWS S3



Amazon S3 (Simple Storage Service) is one of the foundational services in the AWS suite and is widely used by businesses and individuals to store and retrieve any amount of data, at any time, from anywhere.

### Overview of AWS S3:

- **Object Storage:** S3 is an object storage service, meaning it is designed to store unstructured data (like photos, videos, backups, etc.) as objects within resources called "buckets".
- **Durability and Availability:** AWS S3 is designed for 99.999999999% (11 9's) durability over a given year. This ensures that your data remains safe and intact.
- **Scalability:** There's no limit to the amount of data you can store in S3, and it's designed to handle high request rates and traffic.
- **Data Organization:** Data in S3 is organized into buckets (similar to directories) and objects (files).
- **Versioning:** S3 supports versioning, allowing you to retain, retrieve, and restore every version of every object in your bucket.
- **Security:** Offers features like bucket policies, ACLs (Access Control Lists), and server-side encryption (SSE) for data. Integrated with AWS Identity and Access Management (IAM) for access control.
- **Event Configuration:** You can set up event notifications to trigger workflows, alerts, or other automated processes based on changes to your data.

## AWS S3 Pricing Factors

AWS S3 pricing is based on several factors:

- **Storage:** You're billed per GB per month based on the amount of data stored.
- **Requests:** Costs associated with the number and type of requests made (GET, PUT, COPY, etc.).
- **Data Transfer:** While transferring data into S3 is typically free, transferring data out of S3 to the internet or other AWS regions incurs charges.
- **Additional Features:** Features like versioning, monitoring with CloudWatch, data transfer acceleration, and others might have associated costs.
- **Storage Management:** Using features like S3 Inventory, S3 Analytics, and S3 Object Tagging will also influence the total cost.

## S3 command with AWS CLI



The AWS Command Line Interface (CLI) is a powerful tool that allows users to interact with AWS services, including S3, directly from the command line. Here's a list of some commonly used AWS S3 CLI commands:

- **Configuration:**
  - **aws configure:** Setup the CLI with your AWS credentials, default region, and desired output format.
- **Bucket Operations:**
  - **aws s3 ls:** List all buckets.
  - **aws s3 mb s3://my-bucket-name:** Create a new bucket.
  - **aws s3 rb s3://my-bucket-name:** Delete a bucket.
- **File and Folder Operations:**
  - **aws s3 ls s3://my-bucket-name:** List contents of a bucket.
  - **aws s3 cp localfile.txt s3://my-bucket-name/:** Copy a local file to a bucket.
  - **aws s3 cp s3://my-bucket-name/file.txt localfile.txt:** Copy a file from a bucket to the local system.
  - **aws s3 mv localfile.txt s3://my-bucket-name/:** Move a local file to a bucket (removes the local file after copying).
  - **aws s3 rm s3://my-bucket-name/file.txt:** Delete a file from a bucket.

## **Interview Preparation Q&A (Level Wise)**

### **Easy Level**

Q.) What does S3 stand for, and what is its primary purpose?

A.) S3 stands for Simple Storage Service. It is a scalable storage service offered by AWS that allows users to store and retrieve any amount of data at any time from anywhere on the web. It is designed to make web-scale computing easier for developers by providing a simple web services interface.

Q.) How do you create a bucket in AWS S3, and are bucket names globally unique?

A.) To create a bucket in AWS S3, you can use the AWS Management Console, AWS CLI, or SDKs. When creating a bucket, you must provide a unique name that has not been used by anyone else across all of AWS S3 globally. This is because each bucket's name must be unique worldwide, contributing to the DNS system for accessing the bucket.

Q.) What is the default storage class in Amazon S3 when you upload a file?

A.) The default storage class in Amazon S3 when you upload a file is S3 Standard. This class is designed for frequently accessed data, offering high durability, availability, and performance object storage.

Q.) Can you explain what S3 buckets are and how they are used?

A.) S3 buckets are containers for storing objects (files) in Amazon S3. Each object is identified by a unique, user-assigned key. Buckets are used to organise your storage, manage access permissions, and configure settings such as versioning and lifecycle policies. They act as the basic unit of storage in S3 where users can upload, download, and manage data.

Q.) What types of data can be stored in an S3 bucket?

A.) Any type of data in the form of files can be stored in an S3 bucket. This includes documents, images, videos, backups, website static assets, and more. There are no restrictions on file types, making it a versatile solution for a wide range of storage needs.



Q.) How does Amazon S3 ensure the durability and availability of data?

A.) Amazon S3 ensures data durability and availability by automatically replicating data across multiple geographically separated Availability Zones within an AWS Region. S3 is designed to provide 99.999999999% (11 9's) durability and 99.99% availability of objects over a given year, protecting against data loss and outages.

Q.) What is server-side encryption in S3, and how does it work?

A.) Server-side encryption is a method where Amazon S3 encrypts an object before saving it to disk and decrypts it when you access it. S3 handles the encryption/decryption process transparently, providing an additional layer of data security. You can choose to use S3-managed keys (SSE-S3), AWS Key Management Service (KMS) keys (SSE-KMS), or customer-provided keys (SSE-C) for encryption.

Q.) Can you explain the purpose of S3 versioning?

A.) S3 versioning is a feature that enables you to keep multiple versions of an object within the same bucket. This is particularly useful for preservation and recovery purposes, as it allows you to retrieve previous versions of data or restore an object after accidental deletion or overwriting.

Q.) What is the significance of the 'public access settings' for S3 buckets?

A.) The 'public access settings' for S3 buckets determine whether the bucket and its contents can be accessed publicly. AWS blocks public access to buckets by default for security reasons. However, users can modify these settings to enable public access if they need to share files publicly or host static websites. It's crucial to manage these settings carefully to avoid unintended data exposure.

Q.) Explain the concept of a lifecycle policy in Amazon S3.

A.) A lifecycle policy in Amazon S3 is a set of rules that automate the transitioning of objects to different storage classes and manage object expiration. For example, a policy can automatically move objects to a more cost-effective storage class after a certain period or delete old objects that are no longer needed. This helps in optimizing storage costs and managing data retention effectively.

## Medium

Q.) How do you secure sensitive data in S3 at both the object and bucket levels?

A.) Securing sensitive data in S3 involves using a combination of access control mechanisms, including bucket policies, IAM (Identity and Access Management) roles and policies, and S3 ACLs (Access Control Lists) for granular permissions. At the object level, enabling server-side encryption (SSE) for data at rest and using HTTPS for data in transit are key practices. The answer should cover these aspects to demonstrate an understanding of both preventative and reactive security measures in S3.

Q.) Describe the process and benefits of enabling Cross-Region Replication (CRR) on an S3 bucket.

A.) Cross-Region Replication (CRR) is a feature that automatically replicates data from one S3 bucket to another bucket in a different AWS region. The process involves enabling versioning on both the source and destination buckets, and creating a replication rule specifying what objects to replicate and where. Benefits include enhanced data availability and durability, compliance with data residency requirements, and operational improvements such as minimizing latency by serving data from a closer region to users.

Q.) What mechanisms can you use to monitor access and operations performed on S3 buckets and objects?

A.) Monitoring access and operations in S3 can be achieved through AWS CloudTrail logs, which record API calls to your S3 buckets and objects, including the identity of the API caller, the time of the call, the source IP address, and the request parameters. Additionally, S3 server access logging provides detailed records for the requests made to a bucket. These mechanisms are crucial for auditing and understanding access patterns or investigating security incidents.

Q.) Explain the difference between S3 Standard-IA and S3 One Zone-IA. When would you use each?

A.) S3 Standard-IA (Infrequent Access) and S3 One Zone-IA are storage classes designed for data that is accessed less frequently but requires rapid access when needed. The key difference is that S3 Standard-IA stores data redundantly across multiple Availability Zones, offering higher durability and availability, while S3 One Zone-IA stores data in a single AZ, which costs 20% less than Standard-IA but has a higher risk of data loss if the AZ is compromised. Use Standard-IA for critical data that needs high availability, and One Zone-IA for non-critical or replicable data to reduce costs.

Q.) How does S3 Intelligent-Tiering work, and in what scenarios is it most beneficial?

A.) S3 Intelligent-Tiering automatically moves objects between two access tiers based on changing access patterns: a frequent access tier and a lower-cost infrequent access tier. It's

ideal for data with unpredictable access patterns, as it minimizes costs by automatically optimizing storage costs without performance impact or operational overhead. This storage class benefits use cases like data lakes, content distribution, and backup and recovery, where access patterns can vary over time.

Q.) Can you describe how to implement a disaster recovery strategy using Amazon S3?

A.) A disaster recovery strategy in S3 involves leveraging features like Cross-Region Replication (CRR) for geographic redundancy, versioning for recovery from accidental deletions or overwrites, and regular backups to S3 Glacier for long-term storage. The strategy should account for RTO (Recovery Time Objective) and RPO (Recovery Point Objective) requirements, ensuring that critical data is replicated across regions and can be quickly restored in the event of a disaster.

Q.) What is the purpose of S3 Select, and how does it enhance data retrieval?

A.) S3 Select allows users to retrieve only a subset of data from an object by using simple SQL expressions, which can significantly improve performance and reduce costs for data retrieval. It's particularly useful for applications that need to access data stored in S3 without downloading entire files, such as log file analysis or data exploration in large datasets.

Q.) Discuss the importance and application of S3 bucket policies. How do they differ from IAM policies?

A.) S3 bucket policies are JSON-based policies that define access permissions to S3 resources at the bucket level, allowing or denying actions by users, accounts, or other resources. They are directly attached to S3 buckets and can be used to publicly share files, grant cross-account access, or restrict actions to specific IP addresses or VPC endpoints. While IAM policies grant permissions to users or roles within your AWS account, S3 bucket policies control access to the buckets and objects regardless of the user's identity, providing a more granular access control mechanism.

Q.) How can you automate data management tasks in S3, such as cleaning up old files or archiving data?

A.) Automating data management tasks in S3 can be achieved through lifecycle policies. These policies allow you to define rules for automatic transition of objects to different storage classes (e.g., from Standard to Glacier for archiving) or deletion of objects after a specified period. This helps in managing storage costs effectively and ensuring that data is stored in the most cost-effective manner according to its access pattern and relevance.

Q.) What are S3 Access Points, and how do they simplify managing access to shared data sets?

A.) S3 Access Points are named network endpoints attached to buckets that provide customized access permissions and network controls for shared datasets. Each access point supports policies that define a specific access pattern and network configuration, allowing you to manage access at a granular level for different applications or user groups. This simplifies access management for shared data sets by providing a way to segment access according to use case, while improving scalability and security.

## Hard

Q.) How do you implement a strategy for encrypting existing unencrypted S3 objects at scale?

A.) Implementing encryption at scale for existing S3 objects requires a thorough understanding of AWS services and encryption mechanisms. A candidate should discuss using AWS Lambda functions to automate the encryption process by iterating over objects and applying S3's server-side encryption (SSE) using either Amazon S3-managed keys (SSE-S3), AWS Key Management Service (KMS) keys (SSE-KMS), or customer-provided keys (SSE-C). They should also consider the use of S3 Inventory to list all objects and S3 Batch Operations to manage the encryption process efficiently, as well as monitoring and logging to ensure the process's completeness and success.

Q.) Describe a method to optimize costs while ensuring data durability for long-term archival storage in S3.

A.) This question tests knowledge of S3's storage classes and lifecycle policies for cost-effective long-term data storage. The candidate should mention transitioning older data to S3 Glacier or S3 Glacier Deep Archive for reduced storage costs. They should also discuss setting up lifecycle policies to automate the transition of objects to these lower-cost storage classes after a certain period and deleting objects that are no longer needed. Knowledge of retrieval times, costs, and the trade-offs between immediate access versus cost savings in archival scenarios is crucial.

Q.) Explain how to design a secure and scalable access control system for a multi-user S3 environment.

A.) Designing a secure and scalable access control system for S3 involves understanding AWS IAM roles, policies, and S3 bucket policies. A robust solution should include creating IAM roles with least privilege access for different user groups, using bucket policies to enforce fine-grained permissions at the bucket level, and possibly integrating with AWS Organizations for managing policies across multiple AWS accounts. Discussing the use of

S3 Access Points for managing access to shared data sets and the principle of least privilege to minimize security risks would show deep knowledge.

Q.) How would you configure S3 and related AWS services to support disaster recovery (DR) for critical data?

A.) Supporting DR involves configuring S3 with cross-region replication (CRR) to ensure data is duplicated in multiple geographic locations. The candidate should discuss the importance of versioning, CRR, and understanding the RTO (Recovery Time Objective) and RPO (Recovery Point Objective) for their data. Additionally, integrating S3 with AWS CloudFormation for infrastructure as code (IaC) can automate the setup of DR strategies, ensuring quick recovery in different regions. Mention of monitoring with AWS CloudTrail and AWS Config for compliance and governance would demonstrate a comprehensive approach.

Q.) Detail the steps to analyze access patterns and optimize performance for a heavily accessed S3 bucket.

A.) Analyzing access patterns and optimizing performance for an S3 bucket requires leveraging AWS tools and services. The candidate should mention using S3 Analytics to analyze storage access patterns over time, enabling S3 Transfer Acceleration for faster uploads/downloads globally, and considering the use of Amazon CloudFront for caching content closer to users. Discussing the partitioning of objects across multiple buckets or using S3 Access Points to reduce latencies and manage request rates effectively would also indicate a deep understanding of S3's capabilities and performance optimization techniques.

Q.) Explain how to use AWS services to automate data processing workflows involving S3 objects.

A.) Automating data processing workflows with S3 objects involves integrating S3 with AWS Lambda for event-driven processing, Amazon SNS for notifications, and AWS Step Functions for orchestrating complex workflows. The candidate should describe setting up S3 event notifications to trigger Lambda functions for processing data, using SNS topics for alerting and monitoring, and leveraging Step Functions to coordinate multi-step processing tasks that involve reading from and writing to S3, ensuring scalability and flexibility in data processing pipelines.

Q.) Discuss the considerations for using S3 in a hybrid cloud environment, including data synchronization and security.

A.) Using S3 in a hybrid cloud environment requires careful consideration of data synchronization, security, and network connectivity. The candidate should discuss the use of AWS Storage Gateway or AWS DataSync for efficient data synchronization between on-premises environments and S3, emphasizing encryption in transit and at rest, and the management of access controls through IAM and S3 bucket policies. They should also consider the use of AWS Direct Connect for dedicated network connectivity to reduce latency and increase transfer speeds, ensuring secure and efficient data exchange between AWS and on-premises environments.

Q.) How can you ensure compliance with data privacy regulations when storing sensitive information in S3?

A.) Ensuring compliance involves a multifaceted approach including encryption, access controls, and logging. The candidate should talk about encrypting data at rest using S3's server-side encryption options, enforcing strict access controls via IAM policies and bucket policies, and enabling logging and monitoring through AWS CloudTrail and S3 access logs. They should also discuss the importance of regularly reviewing and auditing these controls, as well as potentially leveraging AWS Macie for discovering and protecting sensitive data stored in S3.

Q.) Describe a strategy for managing large-scale data migrations from heterogeneous sources into S3 while minimizing downtime and data loss.

A.) Managing large-scale data migrations requires a strategy that includes pre-migration planning, use of AWS migration tools, and validation post-migration. The candidate should mention conducting a thorough assessment of data sources, utilizing AWS DataSync for online migrations and AWS Snowball for offline data transfer when dealing with large datasets. They should emphasize the importance of incremental data migration strategies, data validation processes to ensure integrity, and minimizing downtime by scheduling migrations during low-usage periods or in phases.

Q.) Explain how to leverage S3 and machine learning services provided by AWS to automate content classification and tagging.

A.) Leveraging S3 with AWS machine learning services for content classification involves using Amazon Rekognition for image and video analysis, Amazon Comprehend for natural language processing, or AWS Lambda functions for custom machine learning models. The candidate should describe setting up S3 event notifications to trigger these services or Lambda functions whenever new content is uploaded, processing the content to classify and tag it automatically, and then updating the object metadata or storing the classification results in a database or another S3 bucket. This showcases an understanding of integrating S3 with AWS's AI/ML services to add intelligence to storage solutions.

## Test your knowledge : Class\_Quiz

Quiz . 15 Total Points

😊 Great job! Keep pushing for that perfect score! 🚀🌟

Your marks : 13 / 15

1. What does S3 stand for in Amazon S3?

- ☐ Simple Storage Solution
- ☐ Simple Secure Storage
- ☐ Secure Storage Service
- ☒ Simple Storage Service

✓ Correct!

2. Which of the following is a valid S3 bucket name?

- ☐ my\_s3-bucket
- ☒ my.s3.bucket
- ☐ my!s3bucket
- ☐ 123-mybucket

✓ Correct!

3. What is the maximum size of an object that can be stored in S3?

- ☐ 5 GB
- ☒ 5 TB
- ☐ 50 TB
- ☐ Unlimited

✓ Correct!

4. Which feature allows you to host a static website on AWS S3?

- ☐ S3 Transfer Acceleration
- ☒ S3 Static Website Hosting
- ☐ AWS Lambda@Edge
- ☐ Amazon CloudFront

✓ Correct!

5. How can you secure data at rest in an S3 bucket?

- ☐ Enable MFA Delete
- ☐ Use VPC Endpoints



5. How can you secure data at rest in an S3 bucket?

- ☐ Enable MFA Delete
- ☐ Use VPC Endpoints
- ☒ Enable Server-Side Encryption
- ☐ Implement IAM Policies

✓ Correct!

6. Which AWS service can be used alongside S3 to distribute content with low latency?

- ☐ AWS Lambda
- ☐ Amazon EC2
- ☐ Amazon RDS
- ☒ Amazon CloudFront

✓ Correct!

7. What is the default storage class in Amazon S3?

- ☒ S3 Standard
- ☐ S3 Intelligent-Tiering

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- ☒ S3 Standard
- ☐ S3 Intelligent-Tiering
- ☐ S3 Glacier
- ☐ S3 One Zone-Infrequent Access

✓ Correct!

8. Which S3 feature should be enabled to track requests for access to the bucket?

- ☐ Access Control Lists (ACLs)
- ☐ Bucket Policies
- ☐ S3 Bucket Logging
- ☒ AWS CloudTrail

✗ Incorrect!

S3 Bucket Logging is the correct answer

9. Which of the following is NOT a feature of Amazon S3?

- ☐ Object Lifecycle Management



9. Which of the following is NOT a feature of Amazon S3?

- ☐ Object Lifecycle Management
- ☒ Provisioned IOPS
- ☐ Cross-Region Replication
- ☐ Versioning

✓ Correct!

10. What does the S3 Intelligent-Tiering storage class do?

- ☐ Encrypts all stored objects
- ☒ Automatically moves objects between two access tiers based on changing access patterns
- ☐ Archives objects to the most cost-effective access tier
- ☐ Replicates objects across multiple AWS Regions

✓ Correct!

11. How can you prevent accidental deletion of objects in an S3 bucket?

- ☒ Enable Versioning
- ☐ Apply a Bucket Policy

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- ☒ Enable Versioning
- ☐ Apply a Bucket Policy
- ☐ Enable MFA Delete
- ☐ Use S3 Glacier Vault Lock

✗ **Incorrect!**  
Enable MFA Delete is the correct answer

12. Which HTTP status code indicates that a successful request was made to an S3 bucket?

- ☒ 200 OK
- ☐ 400 Bad Request
- ☐ 403 Forbidden
- ☐ 500 Internal Server Error

✓ **Correct!**

13. What is the significance of the 's3:ListBucket' permission?

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13. What is the significance of the 's3:ListBucket' permission?

- ☐ It allows users to download objects from the bucket
- ☐ It allows users to upload objects to the bucket
- ☒ It allows users to view the list of objects in the bucket
- ☐ It allows users to delete objects from the bucket

✓ Correct!

14. In S3, what is the purpose of a bucket policy?

- ☐ To configure versioning on the bucket
- ☐ To set up website hosting on the bucket
- ☒ To define access permissions to the bucket and its objects
- ☐ To enable server-side encryption on the bucket

✓ Correct!

15. Which feature allows you to replicate data across multiple AWS accounts?

- ☐ S3 Transfer Acceleration
- ☒ Cross-Region Replication (CRR)



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15. Which feature allows you to replicate data across multiple AWS accounts?

- ☐ S3 Transfer Acceleration
- ☒ Cross-Region Replication (CRR)
- ☐ S3 Batch Operations
- ☐ AWS Organizations

✓ Correct!

Submit